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LDO Input Shaper Toolkit for Klipper

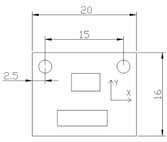
The LDO input shaper toolkit provides all the necessary parts for performing Klipper input shaper calibration with minimal hassle. For detailed information regarding Klipper input shaper functionality see this page . The toolkit contains an easy to mount PCB containing the ADXL345 accelerometer as well as a lightweight FFC ribbon cable and RPI connector board for easy connection between the accelerometer and your Raspberry Pi.

Mounting

A small part can to be printed for mounting the accelerometer PCB onto various toolheads:

- Mini Afterburner Mount - for Voron 0.1
- Afterburner Mount - for Afterburner (Voron 2.4, Trident, Switchwire)
- Stealthburner Mount - for Stealthburner (Voron 2.4, Trident, Switchwire)

If a compatible mount is not available, or if you want to design your own mount, please refer to the following drawing:



Alternatively, you may also choose to mount the accelerometer using double sided tape.

Wiring

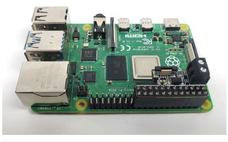
Wiring the input shaper tool is simple, and not intended to be wired permanently. First attach the FFC ribbon cable to each PCB: carefully pull up the black tab on the FFC connector, then insert the FFC ribbon cable and push the tab back in. Take care when pulling up the black tab! It is fragile and **will break** if pulled with excessive force. To make sure the polarity of the ribbon cable is correct, refer to the following picture:

Note: The ribbon cable is NOT to be run in cable chains and should be connected directly for input shaping and removed afterwards.

Check that the blue side of the FFC cable is oriented correctly

Now install the smaller PCB onto your Raspberry Pi as follows:

Mounting Orientation for Version 1 of the ADXL Pi PCB

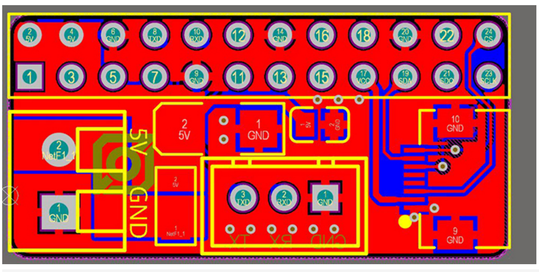


Mounting Orientation for Version 2 of the ADXL Pi PCB

Mounting Orientation for Version 3 of the ADXL PCB on Pi Zero 2W

Changelog

- Version 2 and higher integrates a 5V screw terminal into the Pi PCB. This screw terminal can be used to supply power to the Raspberry Pi instead the USB port (Micro on RPi3 and Type C on RPi4).
- Version 3 of the Pi PCB introduces a serial port connector, which users can optionally use to connect their Pi to a mainboard via serial communication (instead of USB). See the diagram below for detailed pinout.



ADXL PCB circuit diagram of Version 3, a serial port is available in the middle.

Software Installation

Before you can start running input shaper calibration, you need to install a few software packages on your Raspberry Pi. Refer to the software installation section in the klipper input shaper documentation.

Checking the Setup

To test everything works correctly, run the command `ACCELEROMETER_QUERY` in your klipper console. If everything is correct, you should see some measurements from the accelerometer on the console output.

Calibration

The easiest way to perform calibration is to simply run the command `SHAPER_CALIBRATE` in your klipper console. Be sure to read the corresponding section in klipper's documentation for more details and advanced commands and settings.

Comments

No comments yet.

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